

Energy Produced by Human Body

	Walking	Finger typing	Inhalation	Exhalation	Blood flow	Upper limb
Total mechanical energy	67W	7-20 mW	0.8W	1W	0.9W	3W
Electrical energy available	11W	1-3 mW	0.15W	0.2W	0.2W	0.5W
Electrical energy per movement	19 Joules	230-400 microjoules	0.84 Joule	1 Joule	0.16 Joule	2.2 Joules

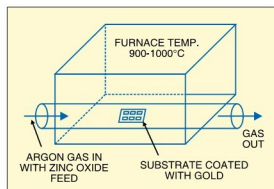


Fig. 1: Zinc oxide, when baked at 900-1200°C in the presence of argon gas, yields wool-like products with nanowires

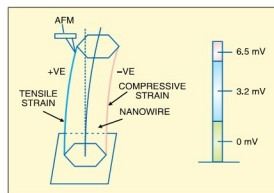


Fig. 2: Piezoelectric effect of bent ZnO nanowire

and audible noises.

Using these tiny power plants, autonomous strain sensors for structures like bridges and environmental sensors for toxin detection could also be run without the need for replacement batteries. They can also supply energy to nanorobotics and microelectromechanical systems (MEMS). MEMS are silicon-based devices having dimensions in the range of microns (millionths of a metre) to millimetres. These are being used in accelerometers of automobile air-bag systems, inkjet printer nozzles and a host of other systems.

Construction

Initially, in 1990s, the researchers constructed a nanogenerator using carbon nanotubes. But it was very difficult to

control the electrical properties of these nanotubes. Work on material science shows that metal oxides are much better at controlling the electrical properties.

Zinc oxide, when baked at 900-1200°C in the presence of argon gas, yields wool-like products with nanowires. Each of these nanowires is a perfect hexagonal-column-like crystal that is grown on a solid conductive substrate using a standard vapour-liquid-solid process in a small furnace.

On a sapphire substrate, gold nanoparticles are deposited to act as a catalyst. An argon gas carrier is flown through the furnace, where zinc oxide powder is heated—all controlled by mass-flow controller (Fig. 1). Zinc oxide diffuses into catalyst particles. The nanowires then grow underneath the gold particles in hexagonal tubular form with continuous supply of feedstock. These are 30 to 100 nanometres in diameter and 1 to 3 microns in length, with average spacing of about 300 nm. Zinc oxide has the rare attribute of possessing both piezoelectric and semiconducting properties.

As shown in Fig. 1, bending back and forth, piezoelectric zinc-oxide nanowires develop a voltage from the compressive and tensile strains on their sides. It is found that when an atomic force microscope (AFM) bends a straight, vertical nanowire, a strain field is established, with the stretched surface showing positive strain and the compressed surface showing negative strain. As the tip of the AFM scans over the nanowires, for each contact position there is voltage varying from 0 to 6.5 mV (Fig. 2). The piezoelectric effect creates an electric field inside

the nanowire's volume, with the stretched and compressed sides of the wire showing positive and negative voltage.

Fig. 3 shows a nanogenerator that consists of an array of paralleled, vertical zinc-oxide nanowires—hexagonal crystals with both piezoelectric and semiconducting properties. All of these continuously generate electricity, which can be collected and delivered to a device. A rectangular electrode with ridged underside sits atop the nanowires and moves side to side in response to external forces such as vibration, human pulse and acoustic waves. Semiconductor nanowires and the conductive rectangular electrode rectify the alternating voltage and release it as direct current. The generator's output is stored in a capacitor and periodically sent to a wireless sensor, which could be measuring blood glucose or pressure. A tiny radio-frequency transmitter reports the results.

The top rectangular electrode is made of platinum-coated silicon with a ridged surface. The ridged side undertakes the function of aligning the ZnO tips. Coating the electrode with platinum both enhances its conductivity and causes it to act like a diode that allows current to flow in only one direction. The electrode is placed above the nanowire array at a controlled distance and can be moved laterally so that it bends the nanowires from side to side.

For minimising the chances of device failure, the packaging technology for assembling the top electrode and the nanowire array is very important. If the electrode presses on the nanowires too firmly, no current will generate. To produce higher voltages, it is necessary to make nanowires with identical height and diameter. Providing suit-